1% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-paraphrased.

Important Notice

The percentage shown reflects a probabilistic assessment and should be treated as an indicator, not a definitive conclusion. Results may vary depending on writing style, subject matter, and document structure. Always use this report as one of several points of reference rather than a sole determining factor.

Guidelines

This report analyzes long-form writing, such as essays, dissertations, and articles. Highlight colors indicate likelihood of AI detection flagging: cyan for high, light purple for moderate, unhighlighted for low.

Non-qualifying content (e.g. bullet points, annotated bibliographies, tables) is not analyzed and may cause the highlighted text to differ from the overall percentage shown. If the document contains non-selectable text, scanned images, or other formatting issues, the score may be inaccurate.

Details

Word Count: 1,786

Document Size: 6.29 MB

Score Interpretation

0% - 19%	Low likelihood of AI-generated content
20% - 50%	Moderate likelihood of AI-generated content
51% - 100%	High likelihood of AI-generated content

Highlight Legend

- High likelihood of being Flagged
- Moderate likelihood of being flagged
- Low likelihood of being flagged

Report generated: September 01, 2026 at 05:50 PM UTC



Inspiring Excellence

BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: Track Alert Vision

by

Group No: 01

Section: 09

Group Members:

Name	ID
Rezowana Mehjabin Lorel	20201089
Ahsan Habib	22201027
Sharan Mistry	22201779
Atahar Shihab	22301336
Maliha Binte Shamim	23101466

Department of Computer Science and Engineering

BRAC University

2/9/26

Abstract

TrackAlert Vision is a robot that aims to reduce the need for continuous human monitoring in controlled indoor environments such as warehouses,laboratories,workshops,storerooms,hallways and offices.It can dodge obstacles,then get back onto its route again to get to the final destination.Furthermore,it can sense human movement,check smoke levels along with gas levels of the smoke and also send live updates to a web page.This report looks at real world consequences this robot might have,where it can actually be used the value it gives,what it cannot do and how it might expand over time.The robot follows a predetermined line to navigate and uses ultrasonic sensing to detect and avoid obstacles.It also employs an MPU6050 to detect tipping or abnormal orientation.A PIR sensor provides initial motion detection,while the onboard ESP32CAM enables camera based verification,image capture and remote monitoring through a web interface.Smoke and gas sensors continuously monitor environmental conditions.

Keywords

Autonomous robotics,linefollowing,IoT,ESP32CAM,obstacle avoidance,environmental monitoring,wireless safety alerts,embedded systems.

1.Introduction

Traditional indoor security and monitoring often depend on human guards,which can be labour intensive,or fixed surveillance cameras,which lack total monitoring capabilities.Thus,TrackAlert can be a low cost monitoring solution with high return value,as it can monitor and detect smoke as well.

The main objective of the project is to build an autonomous robot that can be used as a surveillance system instead of human beings.It can detect and notify of smoke or any kind of movement.

The project aims to:

Automate indoor patrol activities.Detect human movement using an onboard camera.Monitor smoke and gas levels for potential hazards.Detect tipping or abnormal orientation using an IMU.Provide real time sensor information,camera footage and safety alerts through a wireless web interface.It integrates a microcontroller with multiple sensors,actuators and communication modules e.g IR sensors,ultrasonic sensors,PIR sensors,smoke and gas sensors,an MPU6050 IMU,motors and an ESP32 CAM.Sensor inputs are processed by the controller to make movement and safety decisions.And Motor outputs control the robot's navigation.Thus,the project demonstrates practical applications of sensor interfacing,actuator control,embedded programming,wireless communication and real time monitoring.

2.Related Work/ Inspiration

Microsoft has created an indoor Patrol Robot that can record photographs for remote viewing while following predefined routes inside the building. Knightscope, on the other hand develops more sophisticated autonomous security robots that incorporate sonar, LiDAR, thermal imaging and cameras for ongoing surveillance. This strategy is made simpler by our trackAlert vision concept, which emphasizes affordability and practical learning for regulated indoor environments.

3.Technical Approach

Figure 1. Data Flow of TrackAlert Vision

Sensors → Arduino Mega (control) → ESP32-S3-CAM (vision + alerts) → Operator

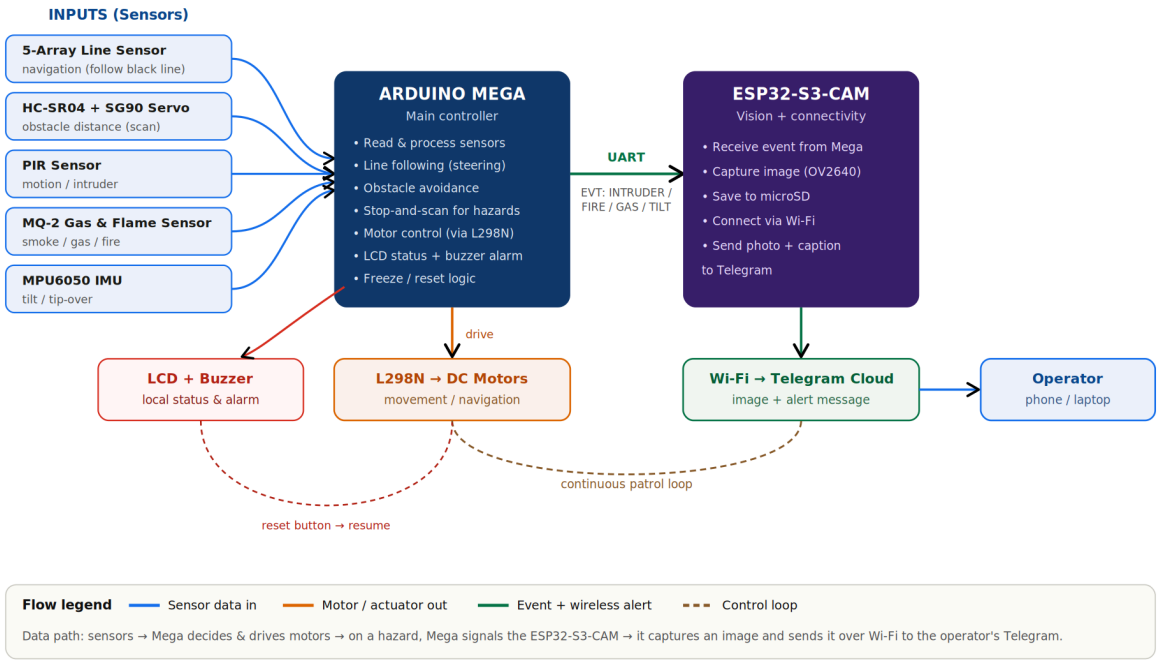
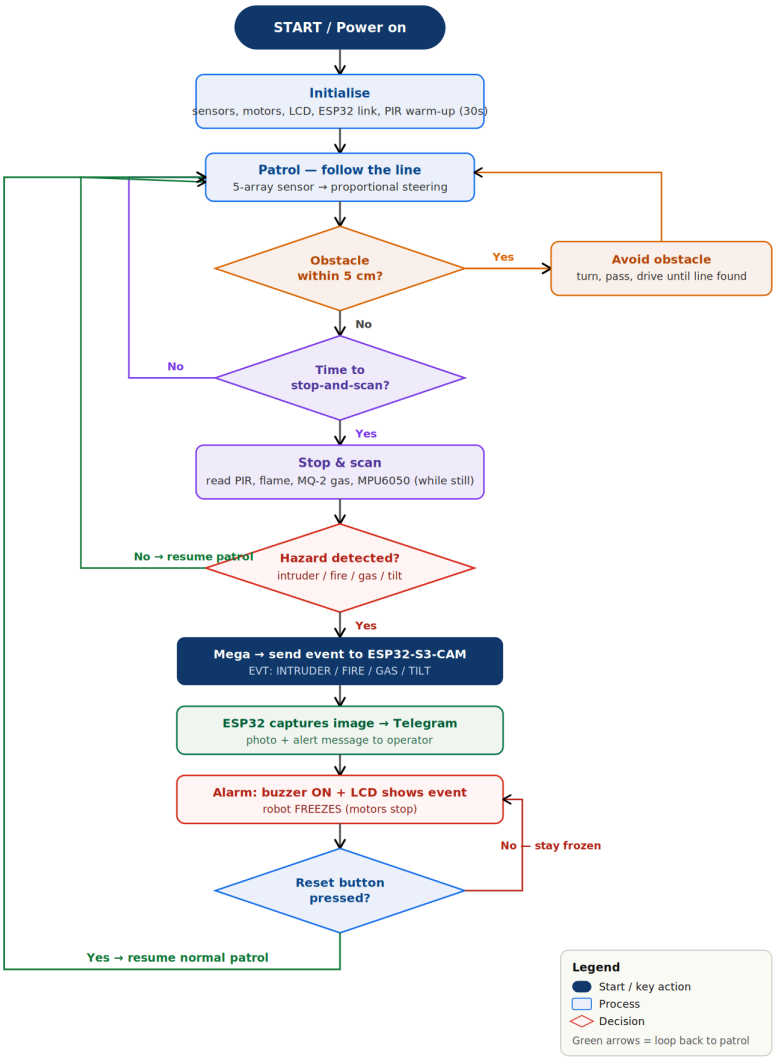


Figure 2. System Flowchart of TrackAlert Vision



Components Used:

Hardware Tools:

ESP32CAM: Main controller responsible for camera operation,image capture and web interface.

IR Line Tracking Sensors: Detect the predefined path and allow the robot to follow the line.

Ultrasonic Sensor: Detects obstacles by measuring the distance between the robot and nearby objects.

MPU6050 IMU: Detects the robot's orientation,tilting and tippingover conditions.

PIR Sensor: Detects movement or human presence around the robot.MQ2 Smoke/Gas Sensor: Detects smoke and potentially harmful combustible gases.

DC Gear Motors: Provide movement

L298N Motor Driver: Controls the speed and direction of the DC motors.Robot Car Chassis and Wheels: Provide the physical structure and mobility of the robot.Battery Power Supply: Provides power to the controller,sensors and motors.Jumper Wires/Breadboard: Used for electrical connections and prototyping.

Software Tools:

Arduino IDE: Used to write,compile and upload the embedded program to the ESP32.

Embedded C/C++: Used to program the robot's movement,sensor processing,obstacle avoidance and safety logic.

ESP32 Web Server: Provides the web based interface for remotely monitoring the robot.

WiFi: Enables wireless communication between the robot and the operator's phone or laptop.

Web Interface: Displays sensor readings,camera feed,robot status,motion events and safety alerts.

Cost Breakdown:

Sl	Items Name	Item Code	Unit Price	Quantity	Total
1	ESP32 S3 CAM Development Board WiFi Bluetooth Module N16R8 High Quality Camera (ESP32-S3)	ROCM-0511	৳1,824.00	1	৳1,824.00
2	LM2596 DC To DC Buck Converter (Step Down Module Power Supply)	ROEM-0269	৳99.00	1	৳99.00
3	**Rocker Switch - ON OFF - 3 Pin**	COSW-0101	৳22.20	1	৳22.20
4	1000uF 50V Capacitor (Electrolytic)	COCA-0320	৳20.00	3	৳60.00
5	2 Wheel Robot Car Chassis Kit	KTCH-0245	৳645.00	1	৳645.00
6	L298N H-Bridge Dual Motor Driver (Red)	ROMT-0314	৳219.12	1	৳219.12
7	Servo Motor Micro SG90 180 Degree Rotation	ROMT-0252	৳135.00	1	৳135.00
8	Line Following Sensor 5 Array	ROSS-0504	৳350.00	1	৳350.00
9	HC-SR501 PIR Motion Detector	ROSS-0196	৳95.20	1	৳95.20
10	MQ-2 Flammable Gas & Smoke Sensor	ROSS-0391	৳150.00	1	৳150.00
11	Flame Sensor Fire Detection Module	ROSS-0332	৳65.00	1	৳65.00
12	LED Red - 5mm	COLD-0113	৳1.68	5	৳8.40
13	LED Blue - 5mm	COLD-0114	৳1.60	5	৳8.00
14	LED Green - 5mm	COLD-0116	৳1.60	5	৳8.00
15	Active Buzzer 5V	ROSS-0254	৳20.00	1	৳20.00
16	Push Button Tactile Switch 4 Pin with Cap	COSW-0434	৳15.00	2	৳30.00
17	220 Ohm 1/4W Resistor	CORS-0201	৳1.99	7	৳13.93
18	2.2k Ohm 1/4W Resistor	CORS-0207	৳2.00	5	৳10.00
19	1k Ohm 1/4W Resistor	CORS-0205	৳1.99	5	৳9.95
20	Jumper Wire for Breadboard 20cm (Female to Female) - Pack of 40	COJW-0080	৳128.00	1	৳128.00
21	Jumper Wire for Breadboard 20cm (Male to Female) - Pack of 40	COJW-0081	৳128.00	1	৳128.00
22	Jumper Wire for Breadboard 20cm (Male to Male) - Pack of 40	COJW-0082	৳128.00	1	৳128.00
23	3.7V 3000mWh Lithium-ion Rechargeable Battery 18650 - A Grade	COPS-0230	৳119.60	4	৳478.40
24	3.7V 18650 Battery Holder (Four Cell)	COPS-0522	৳90.00	1	৳90.00
25	**Universal Lithium Portable Battery Charger (18650 Supported)**	COPS-0244	৳252.00	1	৳252.00
	Item Total				৳4,977.20
	Discount	10%			৳497.72
	Net Payable Amount				৳4,479.48

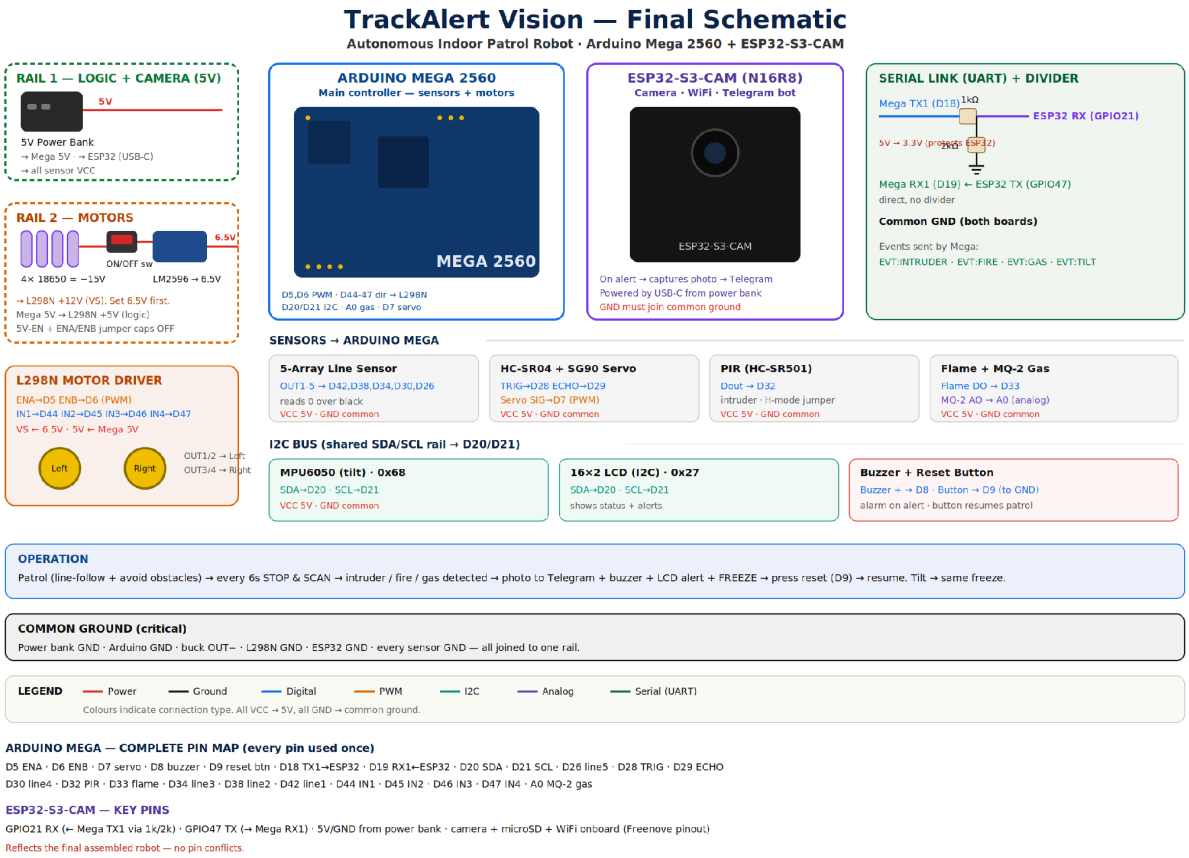


Figure 03: Block diagram of the project.

Functionality:

TrackAlert starts by inspecting the ESP32CAM sensors,motor driver,camera and WiFi connection.The robot then begins patrolling by using IR line following sensors to follow a predefined route.While moving,the ultrasonic sensor continuously checks for obstacles.If any obstacle is detected,the robot avoids it and searches for the line again to resume its route.The MPU6050 is responsible for monitoring the robots orientation and immediately stops the motors if the robot tilts or tips over.Furthermore the PIR sensor detects movement and is verified by the camera before an alert and image are sent to the operator.The smoke sensor continuously monitors the smoke and gas levels and if it crosses a certain threshold,the robot stops and sends a safety alert.All sensor readings,robot status,camera feed and alerts are made available through a WiFi based web interface for remote monitoring.

Challenges:During development,challenges included unstable line following,difficulty recovering the route after obstacle avoidance,false PIR detections,fluctuating smoke/gas readings,robot tipping,limited ESP32CAM processing power and dependence on a physical line.These challenges were addressed through sensor calibration,improved motor control and route recovery logic,camera based motion verification,threshold based safety detection,MPU6050 tilt monitoring,lightweight image processing and careful integration of the hardware and software components.

4.Sustainability & Impact

Sustainability:

TrackAlert Vision a low cost and energy efficient indoor monitoring system.Its line following approach reduces the computational and hardware requirements and thus does not require advanced navigation techniques such as LiDAR and SLAM,making the system simpler.Our system is simple so if any one component malfunctions,we can just replace it instead of replacing the whole robot.The chassis and many other components can be reused for other purposes.

Impact:

TrackAlert Vision is a robot that aims to reduce the need for continuous human monitoring in controlled indoor environments such as warehouses,laboratories,workshops,storerooms,hallways and offices.It can dodge obstacles,then get back onto its route again to get to the final destination.

Future Work:

Future development can include multiple patrol routes,scheduled patrols,industrial grade sensors,LiDAR based navigation places where no physical line exists.We can also implement improved power management and cloud based monitoring.Also,since the ESP32 CAM has limited capability,we can introduce a more powerful camera and integrate AI in our system.

Limitations:

Despite such features which helps in improving human labour,the line following system depends on a physical path.So dirt or any other interruptions to the line can affect navigation.Furthermore,the ultrasonic sensors provide less environmental information than LiDAR.The current system may also have difficulty distinguishing an authorized operator from an intruder.

5.Results & Discussion

The robot was tested in an indoor environment on a black electricaltape track laid over lightcoloured floor tiles.Each subsystem was tested individually and then integrated.The linefollowing,obstacleavoidance and hazarddetection features all functioned correctly after calibration.Representative results are summarised in the table below.

Feature / Test	Result	Notes
Line following (straight + gentle curves)	Working	Stable after reversing sensor order and setting polarity (0 = black)
5 array sensor calibration	Working	All 5 sensors flip 1 (white) / 0 (black) after sensitivity tuning
Obstacle avoidance + line recovery	Working	Detect > turn > pass > drive until line redetected
PIR intruder detection	Working	Reliable at ~1 m after sensitivity tuning; checked during stop and scan

Flame detection	Working	Triggers on a nearby flame (~2050 cm)
Gas/smoke detection (MQ2)	Working	Analogue thresholdbased; requires ~30 s warmup
Tilt detection (MPU6050)	Working	Motors stop when the robot is tipped
Image capture > Telegram	Working	Photo delivered to the Telegram bot on each alert
Alert + buzzer + LCD + freeze/reset	Working	Robot freezes on alert and resumes on reset button press

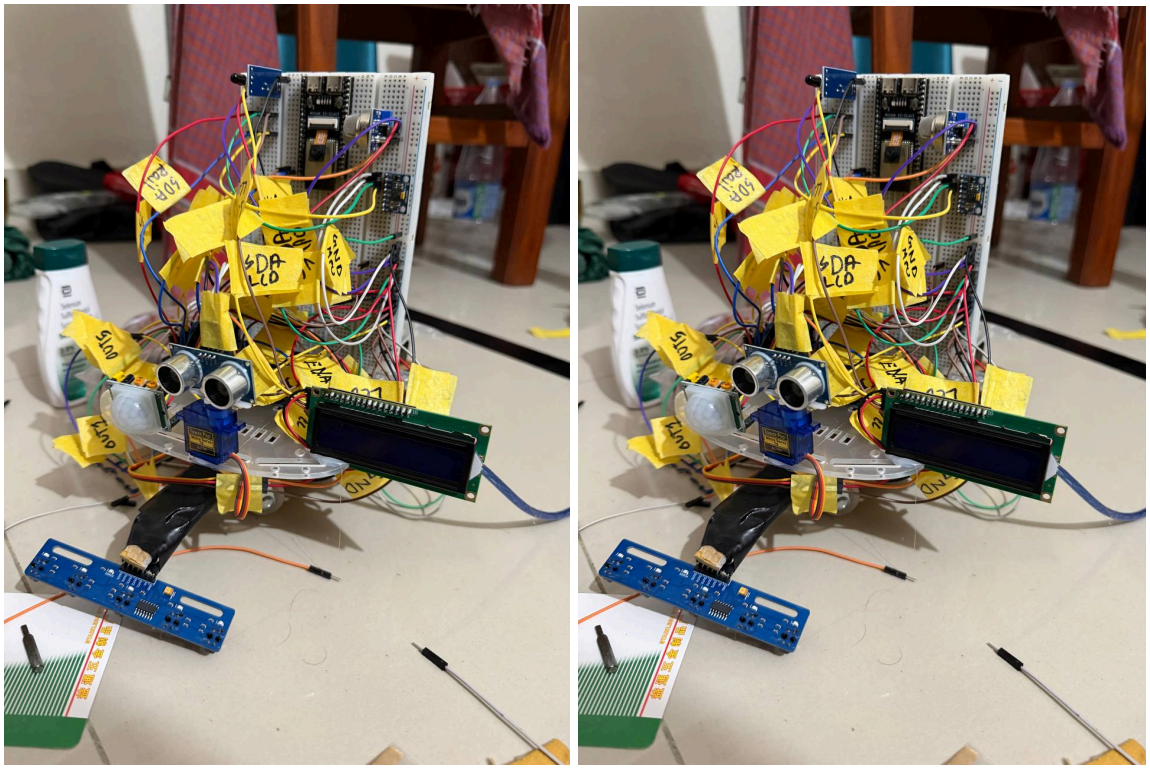


Figure 04: Images of the actual project

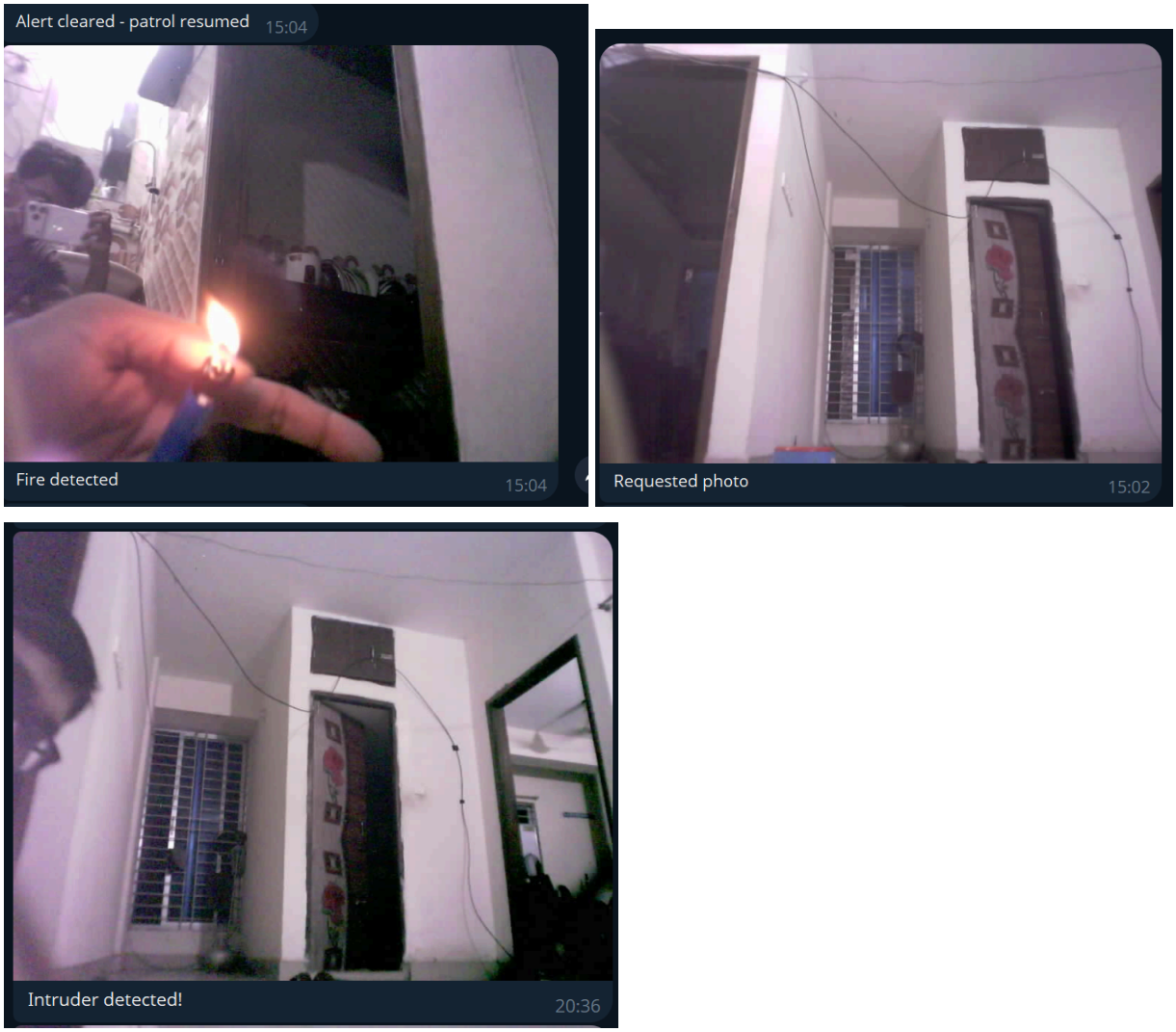


Figure 05: Proof of working outcome.

The results meet the project objectives: the robot patrols autonomously,avoids obstacles and recovers its route,detects intruders and hazards and reports them wirelessly with an image.The main practical difficulties: line stability,route recovery after avoidance and false PIR triggers: were resolved through sensor calibration,sensor driven (rather than purely timed) route recovery and a stop and scan strategy in which the PIR is only read while the robot is stationary.These outcomes confirm that a lowcost,lineguided platform can deliver useful autonomous monitoring without expensive navigation hardware.

6.Conclusion

TrackAlert Vision shows how a low cost autonomous robot can handle the monitoring work,while keeping humans in charge to make decisions that require real judgment.The bottom line idea of the robot,if summarised,is: to MOVE then MONITOR after that DETECT next RESPOND at the end REPORT.Even with its limits the modular setup gives a solid stepping stone for creating a better indoor inspection and observation system.

7.Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Rezowana Mehjabin Lorel	20201089	Hardware Design,Circuit Building & Power Management	Chassis & Power: chassis assembly,L298N motor driver wiring,18650 battery pack,LM2596 buck converter and common ground
Ahsan Habib	22201027	Hardware Design,Embedded Systems & Autonomous Navigation	Line Following: 5 array sensor and proportional steering; Obstacle Avoidance: servo + ultrasonic scanning and route recovery; Telegram Bot: image capture and alert messaging
Sharan Mistry	22201779	Embedded Systems,Wireless Communication & Intruder Detection	Camera & Communication: ESP32S3CAM setup (camera + WiFi),Mega ↔ ESP32 UART communication and PIRbased intruder detection with stop and scan
Atahar Shihab	22301336	Safety Systems,Software Integration & Documentation	Safety & Integration: MPU6050 tilt detection,overall software integration and testing and project report/documentation
Maliha Binte Shamim	23101466	Software Design,Environmental Monitoring & Alert Systems	Hazard Detection & Alerts: flame and MQ2 gas/smoke detection,threshold calibration,buzzer/LCD alerts and freeze until reset logic

References:

Arduino.(n.d.).Obstacle avoiding robot.Arduino Project Hub.Retrieved September 1,2026,from https://projecthub.arduino.cc/aakash11/obstacleavoidingrobot8018ae

Lough,B.(n.d.).Universal Arduino Telegram Bot library: ESP32Cam send photo example [Source code].GitHub.Retrieved September 1,2026,from https://github.com/witnessmenow/UniversalArduinoTelegramBot/blob/master/examples/ESP32/SendPhoto/ESP32Cam/ESP32Cam.ino

Santos,R.,& Santos,S.(2024,June).Telegram: ESP32CAM take and send photo (Arduino IDE).Random Nerd Tutorials.https://randomnerdtutorials.com/telegramesp32camphotoarduino/

Lee.(2024).Building a line follower robot with obstacle avoidance [Video].YouTube.https://www.youtube.com/watch?v=nzZRsMbKrvo

MA Robotic.(2022,December 13).Line follower with obstacle avoiding robot | Arduino and L298 motor driver [Video].YouTube.https://www.youtube.com/watch?v=4PQgjjOqJa4

Techatronic.(2024).Line following robot Arduino tutorial.Techatronic.https://techatronic.com/linefollowingrobotarduinotutorial/